

# Bhaskar De

## Curriculum Vitæ

Postdoctoral Fellow  
Division of Chemical Physics, Lund University  
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### Personal Information

Date of birth: July 22, 1997  
Nationality: Indian  
Work area: Photophysics of Metal Halide Perovskites  
Languages: English, Bengali, Hindi

### Research Interests

- Ultrafast spectroscopy
- Ultrafast laser pulse shaping
- Machine learning for spectroscopy

### Postdoctoral Experiences

July 2026 – present **Lund University**, Sweden  
Postdoctoral Fellow  
Working in the Division of Chemical Physics, Department of Chemistry

### Education

2020 – 2026 **Indian Institute of Science Education and Research (IISER) Bhopal**, India.  
Ph.D. in Physics.  
Thesis Advisor: Dr. Rohan Singh  
Topic: Computational and Experimental Advancements in Optical Two-Dimensional Coherent Spectroscopy

2018 – 2020 **Indian Institute of Technology (IIT) Madras**, India.  
M.Sc. in Physics.  
GPA 8.47/10.0

2015 – 2018 **Ramakrishna Mission Residential College (Autonomous)**, affiliated to University of Calcutta, India.  
B.Sc. (Hons.) in Physics.  
Graduated with First Class Honours

### Publications

- **Bhaskar De**, Pradeep Kumar, Krishna. K. Maurya, Rishabh Tripathi, and Rohan Singh, “Quantitative Lineshape Analysis for Arbitrary Inhomogeneity in Two-Dimensional Coherent Spectroscopy”, *Optics Letters*, **50**, 4502-4505 (2025).
- Pradeep Kumar<sup>†</sup>, **Bhaskar De**<sup>†</sup>, Rishabh Tripathi, and Rohan Singh, “Exciton-exciton interactions: A quantitative comparison between complementary phenomenological models.” *Physical Review B*, **109**, 155423 (2024). (<sup>†</sup> Equal contribution)

- Rishabh Tripathi, Krishna. K. Maurya, Pradeep Kumar, **Bhaskar De**, and Rohan Singh, “**Coherent nonlinear optical response for high-intensity excitation.**” *The Journal of Chemical Physics*, **162**, 114111 (2025).

## Research Experience

- Designed and assembled collinear multidimensional coherent spectroscopy setup, including precise beam alignment, temporal compression, spectrum shaping, and four-wave mixing signal detection
- Gained extensive experience with femtosecond laser amplifiers and optical parametric amplifiers (OPA)
- Developed laser-scanning bright-field reflection microscopy to carry out spatially resolved 2DCS experiments
- Used ultrafast pulse characterization methods (intensity autocorrelation, FROG) to improve time-resolved experiments and achieve sub-60 fs resolution; applied spatial light modulators and grating-based compressors for pulse shaping.
- Applied computational and machine-learning approaches to simulate lineshape simulations, analysis, and modeling exciton–exciton interactions.
- Mentored five Master’s students on projects involving ultrafast optics, experimental alignment, and simulations.

## Experimental Skills

|                                 |                                                                                                                     |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 2DCS Setup                      | Passive phase-stabilized collinear 2D coherent spectroscopy with a 50 kHz pulsed laser; four-wave mixing detection. |
| Pulse Shaping                   | SLM-based spectral/temporal pulse shaping; grating-based dispersion control; vortex beam generation.                |
| Pulse Characterization          | Intensity autocorrelation; SHG-FROG.                                                                                |
| Microscopy & Imaging            | Laser-scanning bright-field reflection microscopy; transmission microscopy.                                         |
| Cryogenics & Optical Techniques | Low-temperature photoluminescence and absorption spectroscopy (down to 4.2 K).                                      |

## Technical Skills

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|--------------------------|----------------------------------------------------------------------|
| Programming Languages    | Python, Fortran, MATLAB, Mathematica                                 |
| Software                 | LabVIEW, Inkscape, Blender                                           |
| Computational Techniques | Parallel Computing, Machine Learning, Deep Learning, Neural Networks |

## Academic Achievements

- 2025 **APS-DLS Travel Award**, APS-Division of Laser Science  
Awarded by the Division of Laser Science of the American Physical Society to support participation in CLEO 2025.
- 2025 **ANRF International Travel Grant**, Anusandhan National Research Foundation, India  
Awarded by the Anusandhan National Research Foundation to support international conference travel to CLEO 2025.

2021 – Present **Prime Minister's Research Fellowship (PMRF)**, IISER Bhopal

Awarded by the Ministry of Education, Government of India, for outstanding research potential in science and engineering. The fellowship provides significant financial support and a research grant for conducting doctoral research.

2022 **PMRF Annual Review Commendation**, IIT Madras

Graded "Recommended with Commendation" during the 2022 PMRF annual review, leading to an invitation to present a highlighted poster at the 2023 PMRF Annual Symposium, IIT Madras.

2020 – 2021 **Junior Research Fellowship (JRF)**, IISER Bhopal

Funded by the Council of Scientific and Industrial Research (CSIR), Government of India. Conducted research in the field of spectroscopy, focusing on two-dimensional coherent spectroscopy.

2018 – 2020 **Competitive Exam Qualifications**

Qualified **GATE**, **JEST**, **CSIR-NET JRF**, and **IIT JAM** – All-India competitive exams for postgraduate and doctoral studies in Physics.

2015 – 2020 **INSPIRE Scholarship**, Government of India

Awarded for ranking in the top 1% of the West Bengal Council of Higher Secondary Education (WBCHSE) Higher Secondary Examination in 2015. This scholarship supports students pursuing higher education in science and engineering.

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## Conference Presentations

May 2025 **Poster**: "Two-Dimensional Coherent Spectroscopy with Arbitrary Inhomogeneous Broadening: Simulations and Analysis." Presented at **CLEO 2025**, Long Beach, CA, United States.

March 2025 **Talk**: "Simulating Two-Dimensional Coherent Spectroscopy with Arbitrary Inhomogeneity Beyond Gaussian Distributions." Presented at **APS Global Physics Summit 2025**, Anaheim, CA, United States.

August 2024 **Poster**: "Two-Dimensional Coherent Spectroscopy Simulations with Arbitrary Inhomogeneous Distribution." Presented at **CLEO Pacific Rim 2024**, Incheon, South Korea.

February 2023 **Highlighted Poster**: "Investigation for the Method of Including Random Inhomogeneity in Simulated 2D Spectra." Presented at the **PMRF Annual Symposium 2023**, IIT Madras, Chennai, India.

November 2022 **Poster**: "Comparative Study of Complementary Many-Body Interaction Models for Two-Dimensional Excitons." Presented at the DAE-BRNS 9th Theme Meeting on Ultrafast Sciences (**UFS 2022**), IISER Thiruvananthapuram, Thiruvananthapuram, India.

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## Teaching Assistantship

IISER Bhopal Waves and Optics, General Physics Laboratory-I, Experimental Techniques, Physics Laboratory II, Basic Electronics

National Programme on Technology Enhanced Learning (NPTEL) 

- NOC22-PH42, Numerical Methods and Simulation Techniques for Scientists and Engineers - 8 Weeks Course
- NOC23-PH02, Physics through Computational Thinking - 8 Weeks Course
- NOC23-PH33, Introduction to Laser - 12 Weeks Course
- NOC24-PH13, Physics through Computational Thinking - 8 Weeks Course

RKMRC ○ "MATLAB BASICS" at RKMRC, Narendrapur, Kolkata  
Narendrapur ○ "Python Program and Applications" at RKMRC, Narendrapur, Kolkata